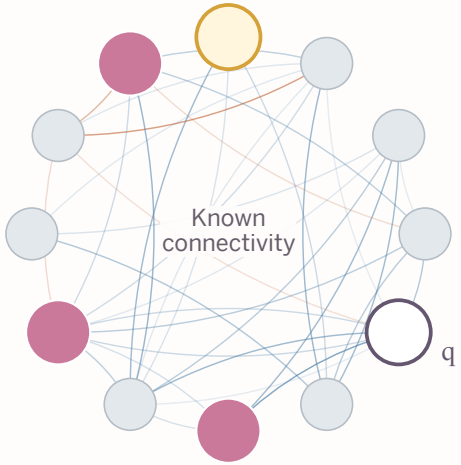
# Predictive risk and recording decisions

## in partially observed neural networks




**12 simulated units · M=3**

Uncertain gains and biases

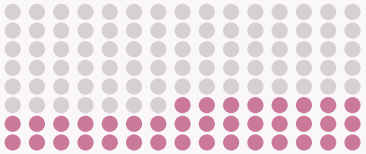
- Recorded node
- Stimulated node
- Response readout (one shown)

### 1 Recording agreement

40 independent networks

**Single-response designs**

**68%** 82 of 120 designs identical recording sets



**Shared three-response designs**

**65%** 26 of 40 designs identical recording sets



same recording set

### 2 Response error

Common MAP fit and budget
Same 40 networks

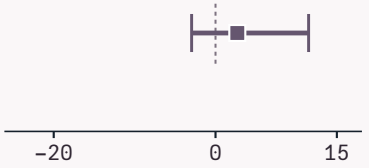
**Single response**

−3.7% [−18.5, +12.1]



**Shared three-response**

+2.7% [−3.0, +11.5]




Mean-MSE change vs nominal (%)

97.5% paired network-cluster CI

### 3 Risk forecasts

Fixed data and point prediction
40 new independent networks

**19** of 40 favour prior averaging; 21 favour nominal

$$S(v,e)=\log v+e^2/v$$

v: MSE forecast; e: response error

One network accounts for 64% of the summed score difference

**Forecast-score difference**

−0.46 [−1.16, +0.02]




Prior-averaged minus nominal

95% paired network-cluster CI

Integrated stimulus response

$$q=\int \Delta r(t)\,dt$$

$$\Delta r(t)=r_{\text{stim}}(t)-r_{\text{control}}(t)$$

We compare nominal and prior-averaged risk across response objectives, using a **common budget and MAP fit**. Identical recording sets and data yield identical predictions; we test forecasts separately on fixed predictions.

Primary error and score intervals include zero. An exploratory extension used 8 independent 100-unit rank-two networks; 4/8 setting-specific intervals exclude zero (unadjusted), all favouring prior averaging; one point estimate favours nominal.